



CST2550 Software Engineering Management and Development
Reset Coursework

Topic: Supermarket Management System

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Section No.**Title**

1	Introduction
2	Data Structures and Algorithms
3	System Design and Database Implementation
4	Features Implementation
5	Testing and Evaluation
6	Conclusion and Future Improvements
7	References

1. Introduction

Small local supermarkets manage many daily activities such as product handling, stock management, supplier tracking, sales recording, and checking product expiry dates. In many small shops, these tasks are often performed manually using paper records or spreadsheets, which can lead to errors such as incorrect stock counts, slow product searching, and difficulties in tracking sales or expired products.

To solve these issues, this project develops a **Local Supermarket Management System**, a web-based application designed to digitalize supermarket operations and improve efficiency. The system helps shop owners store and manage supermarket data in an organized and centralized way, reducing manual work and improving accuracy.

The main operations supported by the system include **user registration and login, product management, supplier management, product search, sales recording, inventory monitoring, stock alerts, expiry monitoring, and revenue calculation**. The system automatically detects low-stock products, highlights products close to expiry, and updates stock quantity after every sale.

The application was developed using **ASP.NET Core Razor Pages, C#, SQL Server, and Entity Framework Core**, providing a practical solution for improving management processes in small supermarket businesses.

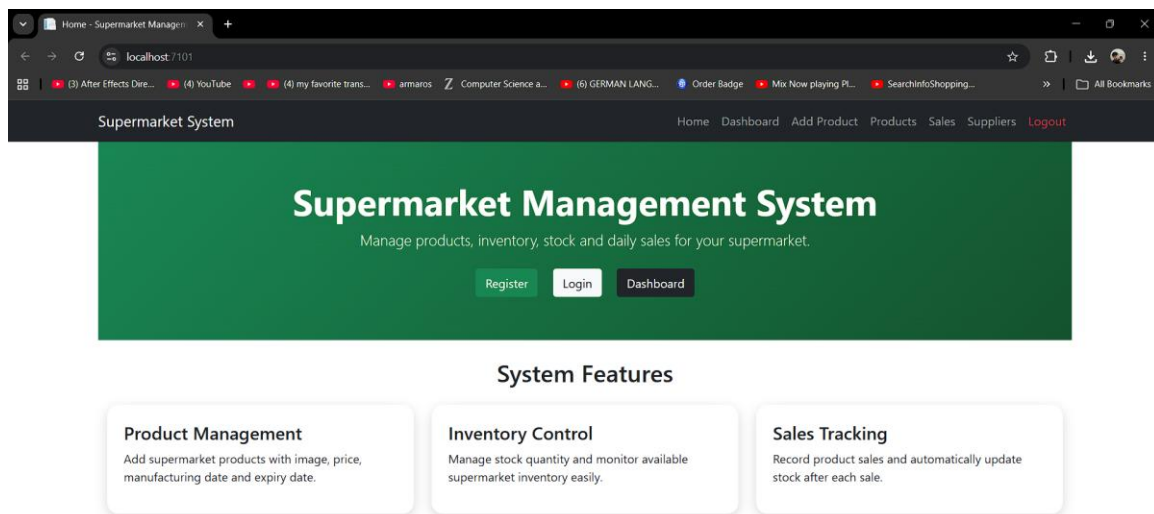


Figure 1: Homepage Interface of the Supermarket Management System

2. Data Structures and Algorithms

To effectively handle goods, suppliers, stock levels, and sales, the Supermarket Management System makes use of data handling techniques. Data must be kept and retrieved fast since supermarket operations need maintaining product details including product name, barcode, quantity, supplier, and price.

The system stores and manages all products, user, and sales data using SQL Server and Entity Framework Core. Products can be swiftly located by using the product name, category, or barcode.

Logic for stock monitoring is also used to track inventory. If the number is less than five, the system indicates Low Stock; if it is less than ten, it signals Reorder Soon. This helps staff identify what needs to be refilled.

The system automatically stores the transaction, determines total income, reduces product quantity, and verifies stock availability when a sale is recorded. Product searching, stock checking, expiry monitoring, and sales computation are the primary methods utilized, which help lower errors and boost supermarket management effectiveness.

Pseudo code:

Search Product Algorithm

```
Start
Input search text
Check product name, category and barcode
Display matching products
End
```

Stock + Expiry Monitoring Algorithm

```
Start
Check product quantity

If quantity < 5
    Display Low Stock
    Highlight panel Red

Else if quantity < 10
    Display Reorder Soon
    Highlight panel Yellow

Check expiry date
```

```

If product expired
    Display Expired

Else if expiry date within 7 days
    Display Expiring Soon

End

```

Sales Recording Algorithm

```

Start
Select product
Enter quantity sold

If quantity sold <= available stock
    Reduce stock quantity
    Calculate total revenue
    Save sales record

Else
    Show Insufficient Stock message

End

```

Time Complexity Analysis

Operation	Description	Time Complexity
Add Product	Product is inserted into SQL database using Entity Framework	O(1) average
Search by Product Name	System checks product names matching search text	O(n)
Search by Barcode	System checks barcode value against product records	O(n)
Update Stock	Product quantity is updated after sale	O(1) after product found
Record Sale	Sale is saved and stock quantity is reduced	O(1) after product found
Low Stock Report	System checks all products and finds quantity below threshold	O(n)
Sales Report	System sums sales records to calculate revenue	O(n)

3. System Design and Database Implementation

The Supermarket Management System was created with SQL Server for data storage and ASP.NET Core Razor Pages with C# backend development. A database is used to store

and manage all supermarket data, while users can interact with the system via a web interface.

To connect the application to the database and carry out tasks like storing, updating, removing, and retrieving records, the project makes use of Entity Framework Core. This increases the effectiveness of database management and lessens the need to write SQL queries by hand.

Three tables are mostly used by the system. Registration and login information is kept in the Users database. Product details, including product name, category, barcode, supplier, price, quantity, expiration date, and image, are kept in the Products table. Details about products sold, amount sold, income, and sale date are all included in the Sales database.

Additionally, the system facilitates CRUD activities, making it simple for users to add, view, update, and remove products. A database-driven solution facilitates supermarket management quicker, decreases manual labor, and increases organization.

Technology	Purpose
ASP.NET Core	Web Development
C#	Backend Logic
SQL Server	Database
Entity Framework	Database Connection

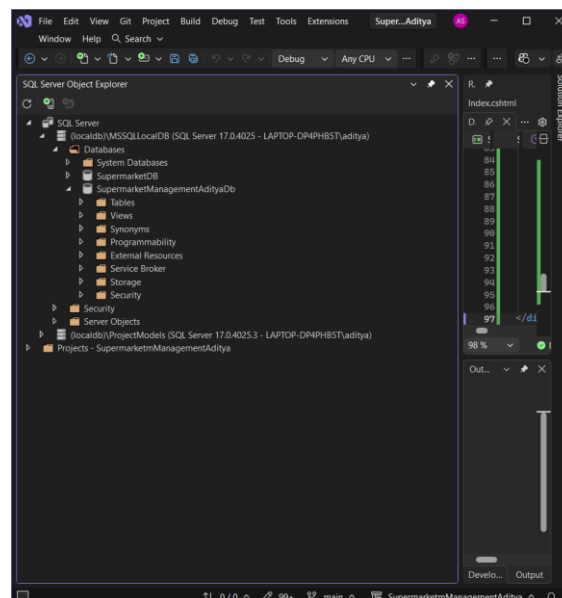


Figure 3: Database Tables Used in the Supermarket Management System

4. Features Implementation

To enhance supermarket operations, a number of features were incorporated into the Supermarket Management System. The technology offers safe access to the dashboard through user registration and login.

By recording information such product name, supplier, barcode, price, quantity, expiration date, and product image, the Product Management feature enables users to add, amend, remove, and search products.

Additionally, the system has stock monitoring, with products below 10 displaying a Reorder Soon caution and those below 5 displaying a Low Stock notice. Expiry monitoring identifies products that have expired or are about to expire.

The Sales feature computes total revenue, automatically lowers stock levels, and keeps track of sold goods. Products are grouped by providers in a provider Dashboard, which facilitates effective supplier information management.

These characteristics decrease manual labor and increase the effectiveness of store management.

5. Testing and Evaluation

The system was tested during development to ensure that all implemented features were working correctly. Different modules such as authentication, product management, supplier management, stock monitoring, expiry monitoring, and sales recording were tested using different input values to verify whether the system produced the correct output. The purpose of testing was to identify errors, improve system reliability, and confirm that the application performs correctly during supermarket operations.

The following test cases were performed during system testing.

Test Case ID	Function Tested	Expected Result	Actual Result	Status
TC01	User Registration	New account created successfully	Account created successfully	Passed
TC02	User Login	User redirected to dashboard	Dashboard opened correctly	Passed

TC03	Add Product	Product stored in database	Product saved successfully	Passed
TC04	Edit Product	Product details updated	Product updated correctly	Passed
TC05	Delete Product	Product removed from system	Product deleted successfully	Passed
TC06	Product Search	Matching product displayed	Search worked correctly	Passed
TC07	Low Stock Check	Warning shown when quantity < 5	Red warning displayed	Passed
TC08	Reorder Check	Warning shown when quantity < 10	Yellow warning displayed	Passed
TC09	Expiry Monitoring	Expired product warning displayed	Expiry alert shown	Passed
TC10	Sales Recording	Stock reduced and revenue updated	Quantity reduced successfully	Passed
TC11	Supplier Update	Supplier details updated	Supplier updated correctly	Passed
TC12	Product Image Upload	Product image stored and displayed	Image uploaded successfully	Passed

The testing process confirmed that all major functions of the supermarket management system were working correctly and the application performed successfully without major errors.

6. Conclusion and Future Improvements

The Supermarket Management System, a web-based tool designed to enhance supermarket operations and minimize manual labor, was developed with success. Product management, supplier management, stock monitoring, expiry checking, and sales recording are just a few of the system's crucial functions that contribute to increased accuracy and efficiency.

This project contributed to the development of practical knowledge in database administration, C#, SQL Server, Entity Framework Core, and ASP.NET Core Razor Pages.

The system can be made more sophisticated and user-friendly in the future by incorporating barcode scanner integration, cloud deployment, PDF report creation, and sophisticated sales analytics.

This project provided practical experience in software development and demonstrated how technology can improve efficiency in real-world business operations.

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